A quadratic lower bound for the second hidden width

Working proof draft

3 November 2026

Proof status. The width theorem below is already verified in Lean for ordinary networks with arbitrary biases. This draft replaces its normalization and probability arguments by simpler proofs. It is not yet a self-contained replacement: the existing finite proof of Lemma 5 has not yet been replaced by a proof in this document. The new normalization is also verified in Lean; the fixed-pair counting argument has an independent mathematical audit and is awaiting its own Lean formalization.

Theorem 1. *For every $C > 0$ there exists $c_C > 0$ such that the following holds for every $N \geq 3$. If an ordinary two-hidden-layer ReLU network with arbitrary real weights and biases represents MAX_N on all of $\mathbb{R}^N$, and its hidden widths satisfy $h_1 \leq CN^2$, then $h_2 \geq c_C N^2$.*

The ordinary architecture is specified in Equation (1); it has no skip connections. The proof uses linear and shallow terms only in intermediate representations. A separate formal theorem records an explicit numerical constant; its optimization plays no role in this draft.

1 Normalization and restriction

1.1 Normalization and coordinate restriction

Write $\sigma(t) = \max\{t, 0\}$ and $\mathbf{1}_I = (1)_{i \in I}$. The network in our theorem has the ordinary form

$$F(x) = b + \sum_{k=1}^{h_2} v_k \sigma\left(c_k + \sum_{j=1}^{h_1} W_{kj} \sigma(b_j + A_j x) \right), \qquad x \in \mathbb{R}^N, \tag{1}$$

with arbitrary real parameters. We assume throughout that $F(x) = \mathrm{MAX}_N(x) = \max_i x_i$ for all x.

Lemma 2. *There is a representation*

$$\mathrm{MAX}_N(x) = Lx + \sum_j B_j \sigma(A_j x) + \sum_k d_k \sigma(g_k(x)), \qquad g_k(x) = U_k x + \sum_j W_{kj} \sigma(A_j x), \tag{2}$$

using at most h_1 first-layer forms and at most h_2 outer ReLU terms, such that

$$A_j \mathbf{1}_{[N]} = U_k \mathbf{1}_{[N]} = 0, \qquad L\mathbf{1}_{[N]} = 1.$$

Proof. For $t > 0$, divide $F(tx)$ by t and let t tend to infinity. Positive homogeneity and continuity of σ give

$$\mathrm{MAX}_N(x) = \sum_k v_k \sigma\left(\sum_j W_{kj} \sigma(A_j x) \right).$$

Thus all biases may be removed without changing either width.

Put $e = \mathbf{1}_{[N]}$, $\alpha_j = A_j e$, and

$$s_k = \sum_j W_{kj}\sigma(\alpha_j), \qquad \ell_k(x) = \sum_{\alpha_j>0} W_{kj}A_j x, \qquad q_k(x) = \ell_k(x) + \sum_{\alpha_j=0} W_{kj}\sigma(A_j x).$$

For fixed x and sufficiently small $\varepsilon > 0$, the kth inner function at $e + \varepsilon x$ equals $s_k + \varepsilon q_k(x)$. The one-sided directional derivative of σ at s in direction u is u if $s > 0$, is zero if $s < 0$, and is $\sigma(u)$ if $s = 0$. Since $\mathrm{MAX}_N(e + \varepsilon x) = 1 + \varepsilon\,\mathrm{MAX}_N(x)$, taking this derivative gives

$$\mathrm{MAX}_N(x) = \sum_{s_k>0} v_k q_k(x) + \sum_{s_k=0} v_k\sigma(q_k(x)).$$

Only forms with $A_j e = 0$ remain inside first-layer ReLU functions. For each retained outer ReLU, $\ell_k(e) = s_k = 0$. Expanding the first sum supplies the linear and shallow terms in Equation (2); its linear part has value $\sum_{s_k>0} v_k s_k = 1$ at e. □

The linear and shallow terms in Equation (2) are introduced by the proof. The statement of the theorem concerns precisely the ordinary architecture in Equation (1).

Lemma 3. *Let $S \subseteq [N]$ be nonempty. A normalized representation as in Equation (2) gives a normalized representation of MAX_S with no larger widths, whose nonlinear first-layer forms are exactly the restrictions of the original forms supported in S.*

Proof. Fix $a \in S$ and define $H_S : \mathbb{R}^S \to \mathbb{R}^N$ by

$$(H_S y)_i = \begin{cases} y_i, & i \in S, \\ y_a, & i \notin S. \end{cases}$$

In particular $H_S\mathbf{1}_S = \mathbf{1}_{[N]}$. Choose $q_i = 0$ for $i \in S$ and $q_i < 0$ otherwise, with $A_j q \neq 0$ whenever $\mathrm{supp}(A_j) \not\subseteq S$. For completeness, enumerate the outside coordinates as $i_1, \ldots, i_r$ and take $q_{i_l} = -t^l$. Each equation to avoid is a nonzero polynomial in t, so all are avoided by taking $t > 0$ sufficiently large.

For every $\varepsilon > 0$,

$$\mathrm{MAX}_N(q + \varepsilon H_S y) = \varepsilon\,\mathrm{MAX}_S(y),$$

because every outside coordinate is less than εy_a. Take the directional derivative at q. A first-layer ReLU survives nonlinearly exactly when $A_j q = 0$, equivalently when its whole support is contained in S. All other first-layer terms become linear or zero. The derivative of the kth inner function is

$$U_k H_S y + \sum_{A_j q>0} W_{kj}A_j H_S y + \sum_{\mathrm{supp}(A_j)\subseteq S} W_{kj}\sigma\big((A_j|_S)y\big).$$

An outer ReLU survives nonlinearly if its preactivation at q is zero; otherwise its derivative contributes only a linear and shallow expression, or zero. This preserves both width bounds. Every resulting inner linear head annihilates $\mathbf{1}_S$, since $H_S\mathbf{1}_S = \mathbf{1}_{[N]}$. The output linear head has value one there, and each surviving first-layer form has zero coordinate sum. Thus the representation is already normalized. □

Lemma 4. *There is an absolute constant $B < \infty$ with the following property. For $n = \lfloor N/2 \rfloor \geq 1$, a normalized representation of MAX_N using at most h_1 first-layer forms and h_2 outer ReLU terms admits a normalized representation of MAX_n with the same width bounds and*

$$\sum_{j:\,|\mathrm{supp}(A_j)|\geq 3} |\mathrm{supp}(A_j)|^4 \leq B h_1.$$

Proof. Choose an n-element subset $S \subseteq [N]$ uniformly. A fixed support T of size $r \le n$ is contained in S with probability

$$\frac{(n)_r}{(N)_r} \le \Big(\frac{n}{N}\Big)^r \le 2^{-r};$$

for $r > n$ the probability is zero. Here $(a)_r = a(a-1)\cdots(a-r+1)$. The constant $B = \sup_{r\ge 3} r^4 2^{-r}$ is finite. Consequently

$$\mathbb{E} \sum_{\substack{j:\,|\operatorname{supp}(A_j)|\ge 3 \\ \operatorname{supp}(A_j)\subseteq S}} |\operatorname{supp}(A_j)|^4 \le B h_1.$$

Choose one S attaining at most this expectation and apply Lemma 3. □

If $h_1 \le CN^2$ and $N \ge 3$, then $N \le 3n$, so the bound in Lemma 4 is at most $9BCn^2$. It controls both the number of supports of size at least three and the sum of their fourth falling factorials: the former is at most one eighty-first of this bound, and the latter is at most the bound itself.

2 The structural ingredient

Lemma 5 (Five-input obstruction)**.** *Suppose a normalized representation of* MAX_5 *has the form in Equation* (2)*, and every nonlinear first-layer function is a scalar multiple of* $\sigma(x_i - x_j)$*. For each second-layer preactivation, combine coefficients and orient the comparisons to write*

$$g_k(x) = \ell_k(x) + \sum_{i<j} a_{k,ij}\sigma(x_i - x_j).$$

Then at least one graph with edges $\{i, j\}$ *for which* $a_{k,ij} \ne 0$ *contains two disjoint edges.*

Remaining proof obligation for this draft. The existing all-real proof uses a star-neuron certificate and a reduction of graphs without two disjoint edges. Its coefficient identities, real parameter coverage, and network interpretation are checked in Lean. A new cycle-deletion proof removes the separate triangle treatment, but retains the star certificate. The present simplification search aims to replace that certificate by explicit identities and a short treatment of their degenerate cases. The arguments below use the established lemma; they do not prove it anew.

3 Counting with disjoint pairs

3.1 A restriction with fixed adjacent pairs

Write $(a)_r = a(a-1)\cdots(a-r+1)$. After normalization, the network has the form

$$F(x) = \ell_0(x) + \sum_{j=1}^{h_1} d_j\sigma(\langle a_j, x\rangle) + \sum_{k=1}^{h_2} v_k\sigma(g_k(x)), \qquad g_k(x) = \ell_k(x) + \sum_{j=1}^{h_1} w_{kj}\sigma(\langle a_j, x\rangle), \tag{3}$$

where all first-layer rows and the inner linear forms ℓ_k, $k \ge 1$, are translation invariant. The output linear form ℓ_0 carries the unit translation slope of MAX. The auxiliary linear forms and shallow output terms in this intermediate model do not change the architecture in the theorem statement. In particular, the original second-layer index set is retained.

Put $r_j = |\operatorname{supp}(a_j)|$ and

$$M = |\{j : r_j \geq 3\}|, \qquad L = \sum_j (r_j)_4, \qquad Q = \sum_{j : r_j \geq 3} r_j^4.$$

Thus $M, L \leq Q$. Every row of support two is a scalar multiple of $e_u - e_v$. Using $\sigma(-t) = \sigma(t) - t$ to give all such rows a fixed orientation, define

$$c_{k,uv} = \sum_{j : \operatorname{supp}(a_j) = \{u,v\}} w_{kj} |a_{j,u}|, \qquad E(G_k) = \{\{u, v\} : c_{k,uv} \neq 0\}.$$

These coefficients are summed before the graph is defined, so signed cancellation between parallel contributions is included.

Fix $n \geq 10$, put $m = n - 5$, and take an integer $1 \leq J \leq m/2$. Choose a uniform permutation of the n coordinate labels. Its first five labels form S; the others occupy ordered tail positions $1, \ldots, m$. The selected coordinates will have value zero. Pair tail positions

$$(1, 2), (3, 4), \ldots, (2J - 1, 2J),$$

and let P be the resulting matching of coordinate labels. Position zero is an anchor with value zero. The gap between positions $r - 1$ and r is $T^{\kappa(r)}$, where κ is a fixed bijection of $\{1, \ldots, m\}$ such that

$$\{\kappa(2), \kappa(4), \ldots, \kappa(2J)\} = \{1, \ldots, J\}.$$

Thus all paired gaps have smaller exponents than every gap between blocks, including the anchor gap. The value at tail position t is $y_t = -\sum_{r \leq t} T^{\kappa(r)}$.

There is one sufficiently large T that works for every sample and every neuron. To see this, each first-row value is a polynomial in T, so its sign eventually stabilizes. After all these finitely many signs stabilize, each second preactivation is also a fixed polynomial. Choose T beyond all roots of the nonzero polynomials. For this common choice,

$$g_k(y) = 0 \quad \Longrightarrow \quad \text{the coefficient vector of } g_k \text{ in the tail variables is zero.} \tag{4}$$

Indeed, if that vector is b, the gap polynomial is $-\sum_r (\sum_{t \geq r} b_t) T^{\kappa(r)}$. Distinct exponents force every suffix sum to vanish, and hence $b = 0$. A first row supported on S is identically zero on the tail subspace and causes no difficulty here.

Declare a first-row support of size at least three to be *pair-bad* if it contains either the endpoints of two pairs in P, or the endpoints of one pair in P and two labels of S. Let B_{pair} be the union of these events. Let B_{selected} be the event that a first-row support of size at least three is wholly contained in S.

Lemma 6. *The two bad events satisfy*

$$\mathbb{P}(B_{\text{selected}}) \leq M \frac{(5)_3}{(n)_3}, \qquad \mathbb{P}(B_{\text{pair}}) \leq \frac{L}{(n)_4} \left[\binom{J}{2} + J \binom{5}{2} \right].$$

Proof. For the first bound, fix three labels in each eligible support. Their probability of belonging to S is $(5)_3/(n)_3$. For the second, there are $\binom{J}{2}$ choices of two pair slots, and $J\binom{5}{2}$ choices of a pair slot and two selected positions. For each choice, all four labels lie in a support of size r with probability $(r)_4/(n)_4$. Sum over choices and rows. □

Lemma 7. *Fix S, the unordered labels in each pair, and all remaining tail labels. If $X_k = |P \cap E(G_k)|$, then*

$$\mathbb{P}\big(g_k(y) = 0,\ B_{\text{pair}}^{\mathsf{c}} \mid \textit{these unordered data}\big) \leq 2^{-X_k}.$$

Proof. The pair orientations are independent fair bits. The bad event is unchanged by these bits, so assume that the fixed data are good.

Consider a zero-sum first-row covector a. Reversing a pair changes only the coefficient of that pair's small gap; all large-gap coefficients are unchanged. If some large-gap coefficient is nonzero, its highest nonzero term determines the sign for our common large T. Thus the activation gate is fixed under every pair reversal.

If every large-gap coefficient is zero, differences of successive block boundary sums show that every singleton tail coefficient is zero, the two coefficients in each pair add to zero, and the selected coefficients add to zero. A row with tail support therefore contains an entire pair with opposite nonzero coefficients. If it has support at least three, it either contains a second such pair, or has a nonzero selected coefficient. In the latter case it has at least two selected coefficients, since their sum is zero. Both alternatives are pair-bad. Consequently, on good data every non-braid row is either supported on S or has a fixed activation gate.

A braid row changes its gate only when its two labels form one of our pairs. It follows that the tail coefficient vector of g_k is

$$b_k + \sum_{i=1}^{J} \varepsilon_i c_{k,u_i v_i}(e_{u_i} - e_{v_i}), \qquad \varepsilon_i \in \{0,1\},$$

with a fixed orientation convention. Its nonzero columns have disjoint supports, hence are linearly independent. At most one assignment to the X_k relevant bits can make this vector zero. Apply Equation (4). □

Lemma 8. *Let G have e edges on $m \geq 2$ vertices, and let P be a uniformly random matching of size J, with $0 \leq 2J \leq m$. Then*

$$\mathbb{E}\, 2^{-|P \cap E(G)|} \leq \exp\left(-\frac{Je}{2m^2}\right).$$

Proof. The cases $e = 0$ or $J = 0$ are immediate. Generate P by a uniform permutation, pairing its first $2J$ positions. For each graph edge ab, swap a into position one and then b into position two. Every permutation with these two fixed entries has exactly $m(m-1)$ preimages, as one sees by undoing the swaps. After forgetting the pair order, the result has the conditional matching law given $ab \in P$.

Put $X = |P \cap E(G)|$ and let X_{ab} be the count after these swaps. At most three pairs change, so $X_{ab} \leq X + 3$. With $F(z) = \mathbb{E}z^X$ and $p = \mathbb{P}(ab \in P) = 2J/[m(m-1)]$, we get

$$F'(z) = p \sum_{ab \in E(G)} \mathbb{E}z^{X_{ab}-1} \geq epz^2 F(z), \qquad 0 < z \leq 1.$$

Integrating $F'/F \geq epz^2$ from $1/2$ to 1 gives $F(1/2) \leq \exp(-7Je/[12m(m-1)])$, which implies the claim. □

Proposition 9. *There is an absolute constant $D > 0$ such that every normalized exact MAX_n network as in Equation (3) satisfies*

$$1 - M\frac{(5)_3}{(n)_3} - \frac{L}{(n)_4}\left[\binom{J}{2} + J\binom{5}{2}\right] \leq \frac{Dh_2}{J^2} \tag{5}$$

for $n \geq 10$ and $1 \leq J \leq (n-5)/2$.

Proof. First fix a second neuron k and put $e = |E(G_k)|$. By a union bound over pairs of disjoint graph edges,

$$\mathbb{P}\big(G_k[S] \text{ contains two disjoint edges}\big) \le \frac{e^2}{2}\frac{(5)_4}{(n)_4}.$$

Conditional on S, the tail pairs form a uniform J-matching on its $m = n - 5$ remaining labels. Deleting S removes at most $5(n-1)$ edges. The preceding two lemmas therefore bound the probability that this neuron is zero, the data are pair-good, and $G_k[S]$ contains two disjoint edges by

$$\frac{e^2}{2}\frac{(5)_4}{(n)_4}\exp\left(-\frac{J(e-5(n-1))}{2m^2}\right). \tag{6}$$

Here the conditional bound is uniform in S; no independence of the selected event and tail graph density is being assumed. The inequality is also valid when $e - 5(n-1) < 0$.

Since $m \ge n/2$ and $J \le m/2$, the positive shift in the exponent is bounded absolutely. Also $(n)_4 \ge n^4/2$. On putting $u = Je/(2m^2)$, Equation (6) is at most an absolute multiple of $J^{-2}u^2e^{-u}$. The latter function of u is bounded, so the neuron cost is at most D/J^2 for an absolute D.

It remains to show that these neuron events cover every good sample. Outside B_{selected}, localization on the five tied coordinates has only braid first-layer rows: genericity says a first-row covector vanishes at the base point exactly when its support is contained in S. The localized network computes MAX_5; its shallow output terms are a linear form and a sum of coordinate-difference ReLUs, as permitted by the structural lemma. By Lemma 5, one of its second preactivations has two disjoint interaction edges. An original second neuron with nonzero preactivation at the base point contributes only linear and shallow first-layer terms, or zero, and supplies no retained outer ReLU. Thus the witness is an original neuron with $g_k(y) = 0$, and its interaction graph on the selected coordinates is exactly $G_k[S]$.

Every sample outside $B_{\text{selected}} \cup B_{\text{pair}}$ is therefore covered by one of the h_2 events just bounded. Sum their probabilities and apply Lemma 6. □

Proof of the existential-constant width bound. Fix $C > 0$ and replace it by $\max\{1, C\}$. By Lemma 4, the original biased network yields an exact normalized model on $n = \lfloor N/2 \rfloor$ inputs, with the same second hidden width and $Q \le ACn^2$ for an absolute A.

Choose a sufficiently small absolute $\delta > 0$, depending only on A, and put $J = \lfloor \delta n/\sqrt{C} \rfloor$. For all sufficiently large n with C fixed, $J \ge \delta n/(2\sqrt{C})$ and $1 \le J \le (n-5)/2$. Since $M, L \le Q$, the two error terms in Equation (5) are respectively $O(C/n)$ and $O(CJ^2/n^2 + CJ/n^2)$. The constants here are absolute. Choose δ so that the quadratic contribution is at most $1/4$, and then take n large enough that the remaining terms have sum at most $1/4$. The counting inequality gives

$$h_2 \ge \frac{J^2}{2D} \ge \frac{\delta^2 n^2}{8DC}.$$

As $n \ge N/3$ for $N \ge 3$, this proves the desired quadratic bound for all sufficiently large N, with C fixed.

In the finite remaining range, $h_2 \ge 1$: an ordinary network with no second-layer neurons has constant output and cannot compute MAX_N. Decrease the positive constant to cover these finitely many values of N. This proves the existence of $c_C > 0$ for every fixed $C > 0$. □

Formal statements. The ordinary-network existential theorem and the separate explicit theorem are in `ordinary/Bounds.lean`. Those existential declarations currently reuse the earlier explicit proof. They do not certify that the new manuscript argument is self-contained.